Comparing Hydra and MS_SCAA Methology

Drafting notes

This document compares the methodology of two Georges Bank multiple species fish population models. Hydra, a length-based multiple species predator-prey population model, and MS_SCAA, an age-based multiple species predator-prey population model.

Introduction

Why develop multiple species fisheries models of Georges Bank

- Movement toward EBFM on Georges Bank
  - The New England Fisheries Management Council, which manages fishing on Georges Bank off the New England coast of USA, is exploring Ecosystem-Based Fishery Management (EBFM), an approach that involves accounting for the species and fisheries in a specific area as a whole, recognizing the energetic limits of the system and taking into account the trophic relationships among species. This could lead to new policies or initiatives that impact multiple fishery management plans or a new fishery ecosystem plan.
- Capture additional dynamics
- See previous papers

Status of Current knowledge

- Two multiple species fisheries model have been developed to evaluate stock status given species interactions on Georges Bank. Hydra, which models species interactions as length based dynamics, and a multiple species statistical catch at age model (MS_SCAA), which models species interactions as age based dynamics.
- Development on the two models to date
- Hydra was developed as a simulation model, and we extend its capabilities to include estimation. MS_SCAA was developed as a 3 species model, that has been extended to a 9 species model. *Additional model status background*

What we add to current knowledge

Having two models allows…
* comparison * strengths and weaknesses * increased confidence due to greater understanding of the impacts of structural uncertainty * combined insights * increased confidence in model predictions * better understanding of the relative role of epistemic vs aleatory uncertainty and structural vs parametric uncertainty

Study Objectives

1. Evaluate these models ability to reproduce simulated population dynamics
2. Compare the performance of the two models ability to reproduce simulated population dynamics
3. Assess the dynamics of Georges Bank populations based on the conclusions of objectives 1 and 2.

Methodology

Study Area and Species

The performance of each model will be assessed for 10 Georges Bank species: Atlantic herring (*Clupea harengus*), Atlantic cod (*Gadus morhua*), goosefish (*Lophius americanus*), haddock (*Melanogrammus aeglefinus*), spiny dogfish (*Squalus acanthias*), winter flounder (*Pseudopleuronectes americanus*), yellowtail flounder (*Limanda ferruginea*), Atlantic mackerel (*Scomber scombrus*), silver hake (*Merluccius bilinearis*), and winter skate (*Leucoraja ocellata*). Later analyses will include analyses with all 10 species in both models but initial analyses will be conducted with the seven species common to both models: Atlantic herring, Atlantic cod, goosefish, spiny dogfish, Atlantic mackerel, silver hake, and winter skate. MS_SCAA was not originally formulated to include all 10 species, with it's original 9 species formulation included pollock (*Pollachius pollachius*), and white hake (*Urophycis tenuis*) which will not be included in the 10 species version developed here.

- The spatial area of the Georges Bank Ecological Production Unit (EPU) is defined as the area shown in (insert figure 1 from this link: https://noaa-edab.github.io/ms-keyrun/allocateLandingsEPU.html). To obtain data for this area available data was collected and partitioned to select only the portion of the data associated with the EPU using the protocol described here: https://noaa-edab.github.io/ms-keyrun/allocateLandingsEPU.html.

MS_SCAA

- See Curti et al and supplement

Hyrda

- See Gaichas et al, Methodology.pdf, and supplement

Comparison

The text below focuses only on the differences between the two models.

Model description

Data requirements

MS_SCAA requires four types of data inputs to run: ecosystem data, species data, species interaction data, and catch data (Table 1). The ecosystem data input is the total biomass of the ecosystem.

- MS_SCAA: Six input data series are required for each species: total commercial catch in weight, aggregate survey index in number-per-tow, age proportions for both commercial catches and survey indices, average individual weight-at-age, and for the multispecies runs, age-specific predator diet.

Table 1: Data Inputs

DataTypes	Descriptions
ecosystem data	total ecosystem biomass
species data	weight at age natural mortality consumption:biomass ratios maximum age of species age of recruitment
species data	age of full recruitment into fishery age of partial recruitment into fishery age of recruitment into survey rate
species interaction data	number of interactions
species interaction data	predator-prey direction
species interaction data	predator species stomach content
species interaction data	consumed species (proportion of stomach weight)
catch data	commercial catch (biomass) commercial catch-at-age (count) survey catch (count) survey catch-at-age (count)

Model decisions: number of length categories

- Hydra: Size bins (number and width [in cm]), Number of fleets, Number of surveys, length to weight conversion parameters, survey biomass, catch biomass, effort, stomach content weight by species and size bin, temperature,

Population dynamics

- MS_SCAA:
- Hydra:

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